Kyle Chong

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EDUCATION

University of Texas at Austin | GPA: 3.47 **Graduation: December 2027**
B.S. Mechanical Engineering

EXPERIENCE

Nanoscale Design & Manufacturing Laboratory — *Undergraduate Researcher* **January 2026 - Present**

- Designed a flexure-based mount for lithography to minimize mask tip/tilt, achieving 155 µrad predicted tilt; validated deflection and stress via FEA against calculated 209 µrad requirement.
- Derived a tolerance stack that summed tilt contributors across the assembly, setting a ±0.015" manufacturing requirement; built components via CNC milling, laser-cutting, and 3D printing (SLA resin).
- Built the LabVIEW-based automation architecture for a novel optomechanical printing system, characterizing 3σ positioning repeatability of 3 µm on a 3-stage precision system (CMOS, mask, aperture).
- Cleanroom-trained in safety, contamination control, and standard gowning/handling procedures.

Imagine Devices — *Associate R&D Medical Device Engineer* **March 2026 - August 2026**

- Designed the analog pressure-measurement system for the company's flagship medical device, quantifying the error budget of ±0.15 cm H_2O now used by the team to measure pressure readings.
- Calibrated the pressure sensor and caught an error in another team's calibration before the DAQ setup was handed off for their R&D testing.
- Plotted sensor data in Python/Excel that exposed high-frequency pump oscillations corrupting sensor readings; designed a pneumatic low-pass filter from damping chambers that removed them from the data.
- Prototyped (SolidWorks CAD) a benchtop neonatal breathing simulator from on-hand parts in place of a ~$10,000 commercial unit; now used as the team's primary testbed.

ENGINEERING PROJECTS

Articulated Wing Robot **May 2026 - Present**

- Integrated a wrist-worn IMU to control a motorized wing, streaming arm angle over BLE to closed-loop motor position control over CAN, achieving 76 ms latency, below the 100 ms real-time target.
- Sized actuator by deriving a quasi-static holding torque of 3.9 N·m per wing (7.8 N·m total) via the Principle of Virtual Work, using SolidWorks mass properties.
- Synthesized a 7-bar pantograph linkage that unfolds a 4.5-ft wing from a single actuator, building a custom Python solver to reduce 135 candidate solutions to the final mechanism.

Snap-Detecting Watch **May 2025 - June 2026**

- Designed a 4-layer flex PCB (KiCad) for a wrist-worn wearable device, integrating an IMU, MEMS microphone, Nordic BLE MCU, and battery power system.
- Built an analog signal chain (op-amp, RC filter, ADC) and Python digital signal processing (DSP) pipeline to train an ML acoustic classifier (F1 = 97.0%)
- Automated data collection with a Python tool, resulting in 2500+ samples passively collected in 2 days.
- Used statistical analysis to evaluate classification models (logistic regression, random forests, and SVM), using 5-fold CV to select random forests as the winning model

COURSEWORK - **Graduate courses italicized*

High-Throughput Nanopatterning	Dynamic Systems & Controls	Statics
Audit: Precision Machine Design	Machine Elements	Dynamics

TECHNICAL SKILLS/CERTIFICATIONS

- Design/Manufacturing: SolidWorks, ANSYS, GD&T, 3D printing, soldering, laser cutting, CNC machining, mill
- Electronics: KiCad, PCB design, Microcontrollers (STM32, Arduino [C, C++]), analog/digital signal processing
- Test & Measurement: LabVIEW, DAQs, oscilloscopes, DMMs, experimental design
- Data Analysis/Visualization: Excel, Python (NumPy, pandas), MATLAB, statistical analysis, Git
- Six Sigma/Lean Yellow Belt Coursera Certification; 8D, Fishbone, DMAIC, 5 Whys, RCCA